2	On a computer motherboard where millions of operations can be performed every second, it uses a variety of capacitors for different operations. For example, 100 pF capacitors are used in high-speed areas to filter electrical noises.	
	(a)	Figure 2.1 shows the variation of the charge stored Q on a 100 pF capacitor with potential difference V applied across the capacitor when charging the capacitor.

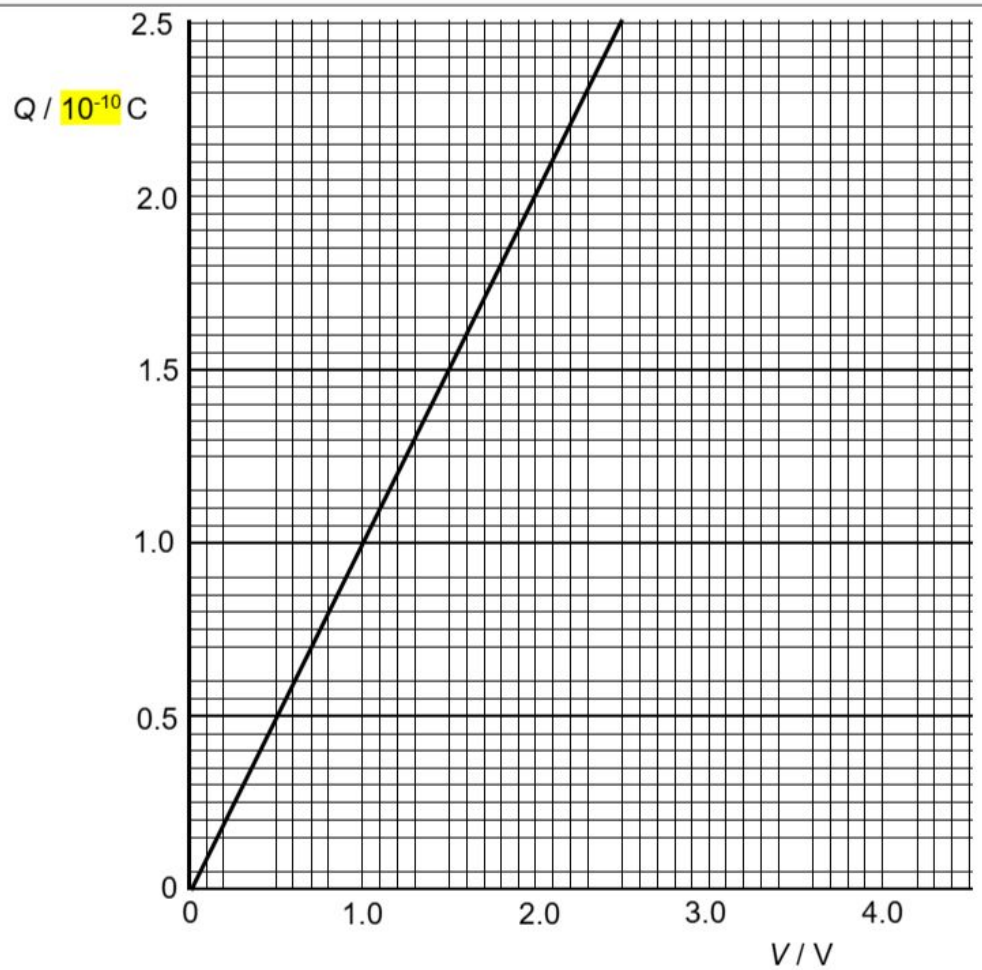


Figure 2.1

(i) Calculate the energy stored in the 100 pF capacitor when the potential difference across it is 2.5 V.

energy stored = ...J [1]

L1 Energy stored = ½ QV
= ½ (2.500 x 10^{-10}) (2.5)
= 3.125 x 10^{-9} J
= 3.13 x 10^{-10} J **A1**

(ii) While smaller capacitors are not able to store enough energy to meet sudden increase in current demands in circuitries, they are excellent at suppressing electrical noises.

On Figure 2.1, draw a line to show the variation of Q with V for a 25 pF capacitor. [1]

L1 $C = \frac{Q}{V}$

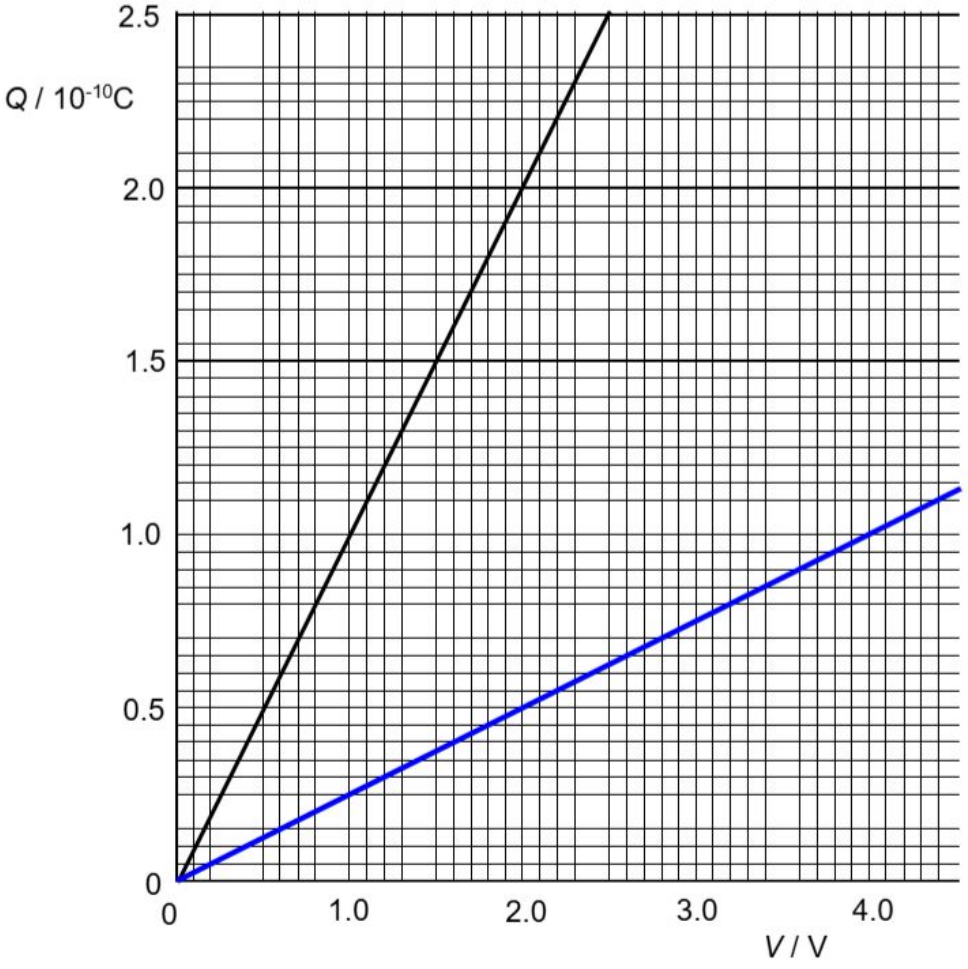
$Q = CV$

∴For a fixed capacitance, graph of Q against V is a straight line

passing through (0, 0) with gradient = *C*.

Since 25 pF capacitance is ¼ of 100 pF, **gradient** of the line for 25 pF capacitor should be **one-quarter** that of the line for 100 pF, **B1**

e.g. when *V* = 2.0 V, *Q* = 0.5 x 10^{-10} C



(b) Figure 2.2 shows six identical 1.0 μF capacitors that are connected in parallel. The capacitors are charged and discharged through the same 10 kΩ resistor. This circuit can be used on the computer motherboard to perform the power-up reset command.

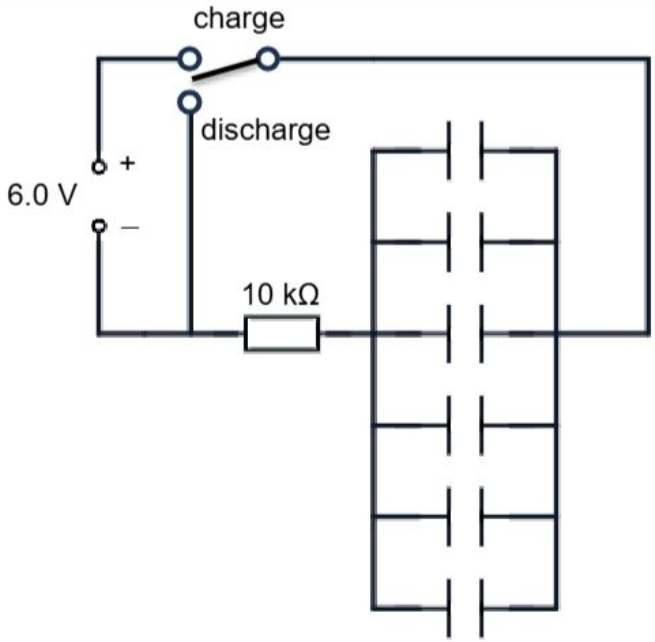


Figure 2.2

The power-up reset command can only be executed when the capacitors are fully charged to 6.0 V, which is equal to the voltage source. When the power-up reset command is executed, the capacitors start to discharge, and the command is completed when the capacitors discharge to 2.0 V or to approximately 33.3% of its maximum voltage V_o. You can take the maximum voltage V_o of the capacitors to be 6.0 V.

(i) Calculate the time taken to complete the power-up reset command.

time taken = ...s [2]

L2

$\tau = RC_e = (10 \text{ x } 10^3)$ (**6 x** 1.0 x 10^{-6}) = 0.060 s

For capacitor that is **dis**charging,

$V_C = V_o e^{-\frac{t}{\tau}}$

$2.0 = 6.0e^{-\frac{t}{0.060}}$ **M1**

$t = 0.065917 = 0.066$ s **A1**

OR

$0.333V_o = V_o e^{-\frac{t}{0.060}}$

$t = 0.065977 = 0.066$ s

		(ii)	To protect the delicate circuits on the motherboard, the circuit is designed such that these capacitors can only start to charge again when its potential difference is equal to or less than 0.50 V. Thereafter, the capacitors charge again until they are ready for the next power-up reset command. Determine the time taken for the capacitors to charge from 0.50 V to full charge. (Assume that a capacitor is fully charged when its potential difference is at 99.3% of its maximum voltage V_o.)	
			time taken = ...s	[3]
		L3	Time taken to **charge** from **0.50 V to full charge (~6.0 V)**: For capacitor **charging**, $V_C = V_o\left(1 - e^{-\frac{t}{\tau}}\right)$ where **time *t* is measured from when V_C = 0**. ∴ Time taken to **charge** from **0.50 V to 6.0 V** =(Time taken to charge from **0 V to 6.0 V**) – (Time taken to charge from **0 V to 0.50 V**) Time taken to **charge** from **0 V to 6.0 V**: $6.0 = 6.0\left(1 - e^{-\frac{t}{0.060}}\right) \rightarrow t$ *is undefined* Assume capacitor is fully charged when V_C = 0.993 V_o, $0.993V_o = V_o\left(1 - e^{-\frac{t}{0.060}}\right)$ **M1** $t = 0.29771$ s Time taken to **charge** from **0 V to 0.50 V**: $0.5 = 6.0\left(1 - e^{-\frac{t}{0.060}}\right)$ **M1** $t = 0.0052207$ s	

			Time taken to **charge** from **0.5 V to 6.0 V** = 0.29771 – 0.0052207 = **0.29249 = <u>0.29 s</u>** *Candidates should show understanding that they need to deduct the time taken to charge to 0.5 V.*	**A1**

[Total: 7]